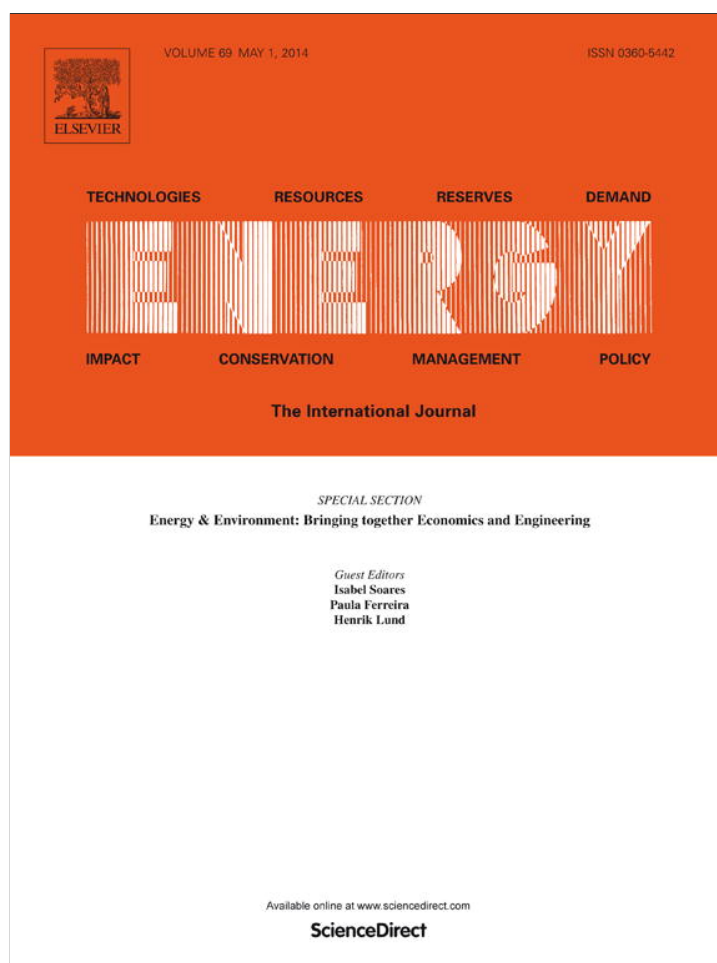




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Assessment of mobility and its impact on energy use and air pollution in Nepal



Sunil Malla*

Technology Consultancy Services, GPO Box 13288, Kalika Marg, Kathmandu, Nepal

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ABSTRACT

Understanding of existing motorized and non-motorized transport services is a key challenge for land-locked Nepal in addressing sustainable transport system development. This paper assesses mobility and estimates current travel demand, energy use and emissions from different transport modes for Kathmandu (capital city) and for rest of the country. Road transport dominates all transport modes in Nepal. In per-capita terms, country's motorized road passenger travel (1461 km) is amongst the lowest in the world. Private vehicles (mainly motor cycle) in Kathmandu and public vehicles (mainly bus) in rest of the country, dominate road passenger travel. Trucks dominate in freight transport services. More than half of country's total commercial energy is consumed by transport sector. Kathmandu alone consumes country's half of gasoline and 20% of diesel supply. The current level of country's road energy use, based on road energy use index, remains one of the lowest in the world. However, emissions of local air pollutant from motor vehicles are significant and they are likely responsible for deteriorating air quality in the country's urban areas. Although less significant in the global context, transport sector is responsible for more than half of country's total energy-related CO₂ emissions.

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1. Introduction

Improvement of accessibility through better transport infrastructure and services is a critical factor for many developing countries' economic and social development. Considering this fact, Nepal government is significantly increasing its spending to develop and maintain transport infrastructure and services. For example, the government budget for transport sector increased from US\$80 million in FY2006/07¹ to US\$596 million in FY2013/14, an increase by 7.5 times [1–3]. In fact, transport is the second of top four sectors (education, transport, health and defense) that received the most government money in last seven years. Despite government's effort, Nepalese people continue to lack adequate and affordable access to transport infrastructure and services. Moreover, in many rural areas of the country, mobility primarily means walking and carrying. For instance, 58% of country's roads are unpaved and most of these roads have earthen surface resulting in reduced accessibility during rainy season [4]. In addition,

Nepalese people walk, on average, 1 h and 14 min to reach the nearest bus stop [5].

Over the past few years, available statistics also show dramatic increase in number of motor vehicles in the country. For instance, number of registered motor vehicles in the country almost quadrupled between 2007 and 2013, driven by the large increase of motor cycles [6]. This rapid increase in number of vehicles resulted in ever increasing demand on imported petroleum fuels putting pressure on foreign currency reserves and raising energy security concerns in the country. For example, between 2007 and 2013, gasoline and diesel import combined increased by 137%, while import value of these fuels almost quadrupled [7,8]. There are also frequent fuel shortages, in part because of Nepal government's inability to pay for fuel imports. Besides, rapidly increasing motorization combined with limited transport infrastructure is causing increasing traffic congestion and rising road accidents, particularly in country's urban areas. Furthermore, public concern over country's deteriorating air quality and the associated health impacts caused by motor vehicles has also grown significantly. Apart from road transport, country's air transport activities also increased rapidly in recent years. For example, number of passengers traveled by air almost doubled between 2006 and 2012, driven by the large increase of both domestic and international passengers [9]. Import of aviation turbine fuel (ATF) also doubled over the same

* Tel.: +977 1 4413040; fax: +977 1 4425219.

E-mail address: malla.sunil@gmail.com.

¹ In Nepal, fiscal year (FY) runs from July 16 through July 15. In this paper, e.g., FY2012/13 is reported as 2013.

Nomenclature

FY	fiscal year
KTM	Kathmandu valley
ROC	rest of the country
GDP	gross domestic product
SRN	strategic road network
LRN	local road network
NH	national highways
FR	feeder roads
MH	mid hill highways
PR	postal roads
UR	urban roads
NRs	Nepalese rupees
ATF	aviation turbine fuel
TJ	tera joule

TD	travel demand
E	energy use
EM	emission (air pollutants)
N	number of registered vehicles
OF	operational factor
VKT	vehicle kilometer traveled
LD	load factor
EI	energy intensity
km	kilometer
pkm	passenger-km
tkm	ton-km
NMT	non motorized transport
ktoe	kilo ton of oil equivalent
l	liter
MWh	mega-watt-hour

period [7]. This clearly indicates that Nepal face three major challenges in transport sector, one is to sustain the supply of fuel and security to meet growing demand for motorized mobility, two is concern about the environmental impact and three is to develop sustainable transport system that is accessible, safe and affordable.

Several research studies and reports have addressed some of these issues either at the national level or at cities level. In particular, concern about deteriorating air quality and increasing local and global pollutant emissions from road transport are frequently highlighted in literature, see e.g., Refs. [10–28]. On the prospects of alternative fuel-based vehicles in the country, several studies [20,29–32] emphasize the importance of electric and hydrogen vehicles. Apart from road transport, ropeways and inland water transport are also important part of mobility in rural Nepal [33–35]. In addition, project reports from several development partners to Nepal find inadequate transport infrastructure and suggest needs for financial and technical support in developing sustainable transport in the country, see e.g., Refs. [36–39]. However, comprehensive assessment of existing passenger and freight transport services by different transport modes for Nepal, including motorized and non-motorized, is lacking in the literature. Furthermore, existing studies on energy and emission analysis of transport sector focused only in Kathmandu, the capital city of Nepal. Nevertheless, assessment of these issues for rest of the country is important as well due to its growing urban population and economic development. Also, existing studies use different data sources resulting in large inconsistencies in estimating energy use and air pollutant emissions from transport sector and the comparison of results has not been seen. Addressing these research gaps is critical for developing country like Nepal in managing future transport demand and in designing energy, environmental and transport policies. From our observation, this is the first kind of detailed assessment of mobility and associated energy consumption and emissions analysis study done for Kathmandu and Nepal.

The contribution from this study is two-folds. First, it assesses the current status of mobility and estimates travel demand, energy use and emission of air pollutants in Kathmandu and rest of the country, with breakdown of mobility, intensity, distance, and fuel mix of all transport modes. Results from this study also are compared with available statistics and research studies. Second, to understand how different countries in the region compare to Nepal, this study analyzes selected road transport indicators and develops road energy use index. This study can provide useful information for policy makers in Nepal and donors involved in country's transport sector development.

2. Overview of Nepal's socio-economic and transport development

Over the last two decades, Nepal witnessed three distinct phases of growth: a phase of high growth during multiparty democratic system (1990–1996), a phase of weakened growth during decade long political conflict (1996–2006) and a phase of high growth in current parliamentary republic system (2006–2012).

2.1. The past and present socio-economic and transportation conditions in Nepal

During 1990–1996, Nepal's gross domestic product (GDP) grew by an average of 5.2% per year while per capita GDP grew by

Table 1
Overview of selected historical socio-economic and transport sector trends for Nepal, 1990–2012 [4,5,6,7,9,40,41,42,43,44,45].

	1990	1996	2006	2012
GDP, PPP (constant 2005, million US\$)	13534	18355	26897	35066
GDP/capita, PPP (constant 2005, US\$)	747	869	1049	1276
Population (total, in millions)	18.1	21.1	25.6	26.9
Urban population (% of total)	8.9	11.4	15.4	17.4
Motorized vehicles (total, in thousands)				
Motor cycles	35	92	355	1032
Cars and jeeps	21	39	83	129
Others ^a	20	45	96	182
Non motorized vehicles (bicycles, in thousands)	1549	3173	4000	6107 ^b
National road network ^c (in kilometer)				
Paved road	2821	3609	5048	10192
Others (gravel and fair-weather road)	3885	7628	12385	14197
Air passenger ^d (in thousands)				
Domestic	291	472	883	1575
International	317	941	1383	2925
Diesel and gasoline sale (in million liters)	118	292	385	848

^a Others include bus, mini bus, micro bus, 3 wheeler, tractor, mini truck, truck, dozer and crane.

^b 2011 value.

^c National road network consists of highways, feeder-, urban-, district-, and village-roads.

^d Figures are based on total passengers in and out from Tribhuvan international airport in Kathmandu.

an average 2.6% per year. These growth rates are comparable to other countries in the region. For example, GDP and GDP per capita during 1990–1996 grew by an annual average of 4.6% and 2.4% in Bangladesh, 5.5% and 3.1% in India, and 5.3% and 4.0% in Sri Lanka [40]. During the same period, country's total population increased by 17% while urban population continued to increase rapidly by 49% mainly due to economic structural transformation and rural to urban migration. The most noteworthy development during this period is the rise in the number of motorized vehicles and air passengers in the country. For example, between 1990 and 1996, the number of motorized vehicles more than doubled. Also, gasoline and diesel consumption combined increased by 2.5 times over the same period, an increase of 16.3% per year (Table 1).

As a result of decade long political instability (1996–2006), growth of country's economy and transport sector slowed down. During this period, country's GDP and GDP per capita fell to 3.9% and 1.5% per year, respectively. Likewise, country's total road length increased only by 4.5% per year during 1996–2006 compared to 9% per year in 1990–1996. Although annual growth of both urban population (5%) and motorized vehicles (11.6%) declined in 1996–2006 period compared to 1990–1996 period, these growth rates remain significant. Also, the annual average growth of gasoline and diesel consumption fell to 2.6% during 1996–2006 mainly due to decline in motor vehicles growth and frequent shut down of vehicular movements in the country.

Despite instability resulting from lingering political uncertainties, the post-conflict period (2006–2012) showed rise in growth of GDP (4.6% per year) and GDP per capita (2.7% per year). However, during this period, motorized vehicles and road length increased substantially both in absolute values and in growth rates. For instance, the number of motorized vehicles almost tripled from 534 thousand in 2006 to 1518 thousand in 2012, 19% per year, while the road length increased from 17 thousand km in 2006 to 24 thousand km in 2012, 6% per year. Likewise, diesel and gasoline use continued to increase rapidly by 14% per year during 2006–2012.

Apart from road transport, air transport also plays a vital role in country's transport system. It provides services to remote areas which are inaccessible by roads and contributes to tourism development which is an important economic activity in the country. For example, in 2012, travel and tourism's total contribution in the economy is estimated to be NRs 147 billion, 9.4% of GDP, and this industry provided 1256 thousand jobs, about 8.2% of total employment [46]. The air traffic volume also increased considerably over the past 20 years. For example, between 1990 and 2012, the number of international passengers increased from 317 thousand to 2925 thousand, growth of 10.6% per year, while the number of domestic passengers increased from 291 thousand to 1575 thousand, growth of 9.4% per year (Table 1). In fact, from 2006 to 2012, the number of international air passengers grew by 13% per year, mainly due to increase in tourists and the movement of number of migrant workers traveling overseas.

As mentioned in the previous section, Kathmandu valley (KTM), which includes Kathmandu, Lalitpur and Bhaktapur districts, also plays an important role in country's transport system development. The valley is the main economic hub and the largest urban center of Nepal. In 2011, about one-third of country's total value of economic activities is estimated to be concentrated in KTM [47]. In 2012, two-thirds of country's personal vehicles, mainly car and jeep, and one-third of country's motor cycles are running in KTM [6]. In KTM, between 2002 and 2012, available statistics show passenger cars more than doubled and motor cycles more than tripled (Fig. 1), while road length increased by only 67% [48]. As a result of this, KTM's traffic congestion is increasing costing money in wasted fuel and time. Moreover, deteriorating road safety is leading to rising traffic accidents. For example, between 2002 and 2012, road

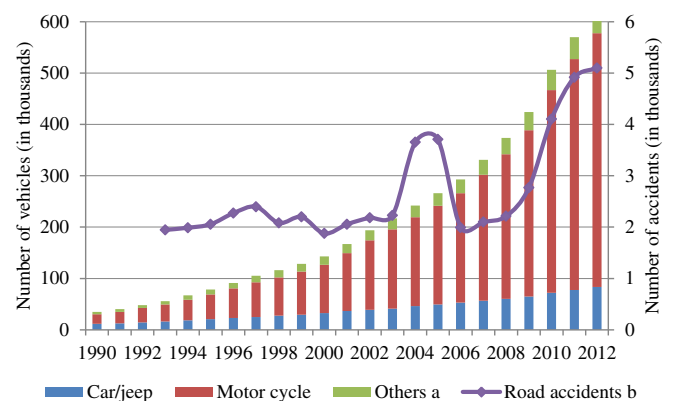


Fig. 1. Number of motorized vehicles and road accidents in KTM, 1990–2012 [6,49].
^a Includes remaining motorized vehicles. ^b Road accidents involve injury to people including fatalities and damage to property.

accidents and traffic related deaths in KTM increased by 134% and 97%, respectively [49].

2.2. The current status of road networks in Nepal

The importance of roads often recognized and highlighted in key national plans and strategies. Nepal government's 13th plan (2014–2016) has emphasized the role of roads in reducing poverty in the country [50]. The plan aims to connect all 75 district head quarters through roads. Besides roads, the plan also aims to construct two international airports and to carry out activities necessary for the development of national east–west long distance railways and metro railways in KTM. Selected key transport related government strategies at the national level include 2001 road transport policy for the development of 20 year road master plan, integrated 10 year sector wide plan and a priority investment plan (2007–2016) for the development and management of strategic roads, 2004 local infrastructure development (LID) policy based on the national strategy of rural infrastructure development (1997) and establishment of roads board Nepal (RBN) in 2003. Likewise, at the city level, selected key strategies include Kathmandu valley urban road development program and Kathmandu tarai road fast tract project.

In Nepal, national road network consists of Strategic Road Network (SRN) and Local Road Network (LRN). SRN is managed at the central level by the Ministry of Physical Planning and Works and this network consists of national highways (NH), feeder roads (FR), mid hill highways (MH), postal roads (PR) and strategically important urban roads (UR). LRN is managed at the local level by the Ministry of Local Development and this network consists of urban roads that are not included in SRN and district or village roads. In 2012, Nepal's SRN is estimated at 11636 km in length and this network connected 73 of country's 75 districts [48]. Half of these roads are paved roads and more than one-third are seasonal dirt roads (Table 2). About 57% of country's SRN is FR, followed by NH (29%), MH (9%) and PR (5%) (Table 3).

However, the length of SRN both by road class and by road surface at the regional² level varies widely due to country's diverse terrain and geography. For example, 55% of all SRN is in hills, followed by tarai (34%) and mountains (11%). About 59% of SRN in tarai is paved roads while it is only 44% in hills and 32% in mountains. Region-wise, 27% of all SRN is in central development region, followed by eastern (22%), mid-western (20%), western (19%) and far-

² Nepal is divided into 3 distinct east-west ecological regions: mountains in the north, tarai in the south, and hills in between. Administratively, country is divided into 5 development regions: eastern, central, western, mid-western and far-western.

Table 2

Strategic and local road networks by ecological and administrative region with road surface in Nepal, 2012 [44,48].

Region	Strategic road network (SRN)				Local road network (LRN)			
	Total	% Share in total			Total	% Share in total		
	(km)	Paved	Gravel	Earthen ^a	(km)	Paved	Gravel	Earthen ^a
Mountains	1231	32	8	59	4264	0	4	96
Eastern	218	35	11	53	799	0	2	98
Central	399	67	7	26	2606	0	5	95
Western	211	0	0	100	301	0	0	100
Mid-western	202	0	0	100	348	0	0	100
Far-western	201	27	25	48	210	0	0	100
Hills	6447	44	11	45	25218	3	8	89
Eastern	1356	31	8	60	4214	1	7	93
Central	1671	58	16	26	8327	5	18	77
Western	1449	42	5	53	9928	2	3	95
Mid-western	1302	33	20	47	1936	0	2	98
Far-western	668	63	1	36	814	0	1	99
Tarai	3958	59	27	14	21462	4	57	38
Eastern	995	66	22	13	4423	3	62	35
Central	1103	54	31	15	7817	3	43	54
Western	591	72	16	12	4053	11	51	38
Mid-western	804	55	31	13	3310	2	71	27
Far-western	465	46	35	19	1858	0	99	1
Nepal	11636	48	16	36	50943	3	29	68

^a Earthen road in SRN includes motor vehicles passable dirt roads while earthen road in LRN includes all dirt roads including trails and tracks.

western (11%) regions. Moreover, about one-third of all paved roads are in central region, followed by eastern (21%), western (19%), mid-western (16%) and far-western (12%) regions. Almost two-thirds of NH, FR and PR combined are concentrated in eastern, central and western regions. The road density of SRN is 7.9 km per 100 square km and 0.44 km per 1000 people.

In 2012, the total length of LRN is estimated at 50943 km with 6683 number of roads [44]. More than two-thirds of these roads are dirt roads, while only about 3% are paved roads and 29% are gravel roads (Table 2). Furthermore, most of these dirt roads are seasonal roads and these roads include motor impassable trail and track roads. Half of these roads are in hills, followed by 42% in tarai and the remaining 8% in mountains. Over 95% of LRN in the mountains and close to 90% of LRN in the hills are dirt roads, while it is only 38% in tarai. The road density of LRN is 35 km per 100 square km and 1.9 km per 1000 people.

The total length of SRN for KTM is estimated at 473 km (Table 4). It is interesting to note that only two-thirds of these roads are paved roads and 69% of these roads are major FR. In contrast, the length of LRN for KTM is estimated at 1746 km. About 58% of these roads are graveled roads and 26% are earthen roads while the remaining 16% are paved roads.

3. Methodology and data

3.1. Methodology

Over the years, many modeling techniques have evolved in travel demand forecasting and their application for solving

Table 3

Strategic road network (SRN) by development region with road class in Nepal, 2012 (km) [48].

Region	National highway	Feeder road (major)	Feeder road (minor)	Mid hill road	Postal road
Eastern	797	1180	28	429	136
Central	876	1888	137	61	212
Western	478	1385	194	142	51
Mid-western	735	1948	216	309	100
Far-western	525	627	0	87	94
Nepal	3411	6029	575	1028	592

common transport problems, including conventional “four-step” model and advanced activity based-, dynamic traffic assignment- and traffic simulation-models [51,52]. Likewise, works on building theoretical framework for modeling the interactions between energy and transport service demand have gained popularity in recent years [53]. However, many of these sophisticated transport demand modeling techniques and their application require detailed reliable data often not available in developing countries. For example, there are no published data on how far vehicles by type and fuel are driven and how much fuel each vehicle–fuel combination consumes in Nepal. For simplicity, we follow commonly used activity–structure–intensity–fuel (ASIF) framework using bottom up approach in estimating road travel demand and its associated energy use and air pollutant emissions [54,55]. In this accounting framework, travel demand is a function of mode, technology choice, total distance traveled, driving style and vehicle occupancy, and transport energy use is a function of travel demand and fuel efficiency. Emission of air pollutants in transport sector is the product of level of activity (passenger and freight travel), structure (share by mode and vehicle type), fuel intensity (fuel efficiency) and emission factor. In more sophisticated formulations, life-cycle energy use and emissions analysis of transport sector considers vehicle production, infrastructure provision and fuel production [56].

Broadly, Nepal's transport sector is classified into road, railways, air, inland water and ropeways (Fig. 2). Motorized and non-motorized roads are the dominant modes of transport in the country. While air transport provides limited services to key commercial and tourist areas, services from railways, ropeways and inland water transport are minimal. In fact, the transport services from both passenger railways and freight ropeways in the country

Table 4

Road network by road class with road surface in Kathmandu valley, 2012 (km) [44,48].

	Paved	Gravel	Earthen
Strategic road network (SRN)	312	85	76
National highways	61	11	0
Feeder road (major)	219	55	53
Feeder road (minor)	32	18	23
Local road network (LRN)	286	1012	448

stopped operating for over a decade. This study considers only road and air transport. Based on road and air transport classification (Fig. 2), passenger from freight transport is separated and distinguished between each transport modes. For road transport, travel demand (TD), energy use (E), local air pollutant emissions (EM) and CO₂ emission for road passenger- and freight-transport are estimated for KTM and ROC using the following equations:

$$TD_i = \sum_j \sum_k N_{j,k} \cdot OF_{j,k} \cdot VKT_{j,k} \cdot LF_{i,j,k} \quad (1)$$

$$E_i = \sum_j \sum_k N_{j,k} \cdot OF_{j,k} \cdot VKT_{j,k} \cdot EI_{i,j,k} \quad (2)$$

$$EM_{i,l} = \sum_j \sum_k N_{j,k} \cdot OF_{j,k} \cdot VKT_{j,k} \cdot EF_{i,j,k,l} \quad (3)$$

$$CO_{2ij} = \sum_k E_{i,j,k} \cdot EF_k \quad (4)$$

where N is the number of registered vehicles, OF is the operational factor (in fraction), VKT is the vehicle km traveled (km), LF is the load factor (occupancy in passenger or ton), EI is the energy intensity (l/km or kg/km or MWh/km) and EF is the emission factor (g/km or kg/GJ). The subscript i is the transport type, j is the transport mode, k is the energy type and l is the local air pollutant type. The following local air pollutant emissions are considered: carbon monoxide (CO), hydrocarbons (HCs), sulfur dioxide (SO₂), particulate matter (PM₁₀), oxides of nitrogen (NO_x) and poly aromatic hydrocarbons (PAH). CO₂ emission is directly related to the amount of fuel used, while emissions of local pollutants depend on the amount of fuel used and other location specific factors such as speed, acceleration and load on the vehicle, vehicle technology and fuel type. Road TD is measured in passenger-km (pkm) for passenger and ton-km (tkm) for freight transport. For air transport, TD, measured in number of passengers for passenger transport and in ton for freight transport, and E, energy use, are compiled from published statistics. CO₂ emission is then estimated based on energy use and IPCC's (Intergovernmental Panel on Climate Change) tier-1 emission factor [57]. Current analyses of TD, E and EM are carried out using excel spreadsheets.

3.2. Data

The above mentioned approach requires gathering of detailed statistics including vehicle stocks, vehicle ownership and surveys that assess transport characteristics such as operational factors, load factors, distance traveled, vehicles fuel economy and emission factors. For the purpose of this study, none of the available literature provides complete dataset. Data used in estimating TD, E and EM are based on various sources and assumptions (Table A.1). While these sources report data on different energy sources in different measurement units, all values are converted to the common energy units of kilo ton of oil equivalent (ktoe) using caloric and other conversion factors. Likewise, emission factor used in estimating EM for motorized road transport is provided in Table A.2. Due to lack of country-specific studies, local air pollutant emission factors are taken from existing studies in India because of similarities in vehicle type, conditions, and fuel economy. CO₂ emission factors by fuel type are based on IPCC's tier 1 default values that assume 100% of the carbon present in the fuel is oxidized during the combustion process. In addition, population and number of households used in the study are extrapolated based on national population and housing census 2011 and the past census data published by Central Bureau of Statistics (CBS).

4. Results and discussion

In this section, travel demand and associated energy use and emission of air pollutants in 2013 are discussed for KTM and ROC. The primary focus of this analysis is in modal share of travel and freight demand in the country.

4.1. Travel demand analysis

In 2013, total road passenger travel demand in Nepal is estimated at 54.7 billion pkm, of which 18% comes from KTM alone (Table 5). Three-fourth of this demand is met by motorized vehicles and the remaining one-fourth is met by NMT (bicycle and walk). Excluding NMT, road passenger travel demand in KTM is estimated at 7.9 billion pkm (20%) in 2013, while it is estimated at 32.1 billion pkm (80%) in ROC. In per capita terms (pkm per person), Nepal's motorized road passenger travel of 1461 in 2013, is one of the lowest, when compared to Pakistan (1547 in 2005), S. Korea (2046

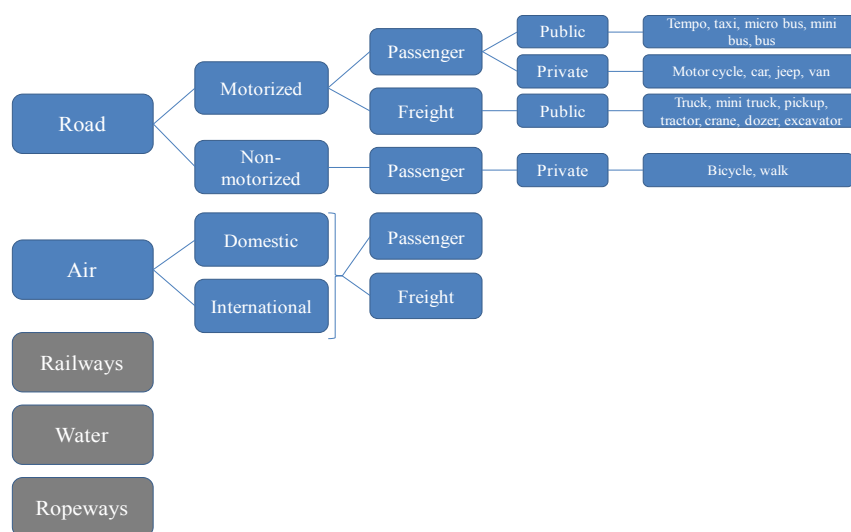


Fig. 2. Transport sector classification in Nepal.

in 2009), India (4592 in 2011), Japan (7094 in 2008), UK (11907 in 2009), Australia (13736 in 2009) and the USA (25894 in 2008) [40]. These figures reflect, among others, the relative importance of passenger road transport in the country.

Although NMT is quite significant both in KTM and ROC, it translates into a much smaller share of total pkm traveled. For example, in 2013, traveling on foot and bicycle represents about 20% and 7% of total pkm traveled, respectively. At the regional level, traveling on foot and bicycle combined in KTM represents only about 2% of total pkm traveled, while it is much higher, at 29%, in the case of ROC. As KTM is relatively a small city, commuting to places can be easily reached on foot or by bicycle. This suggests that the share of NMT peak-hour travel demand in KTM is most likely significant. In fact, few studies reported the share of NMT in peak-hour travel demand in KTM ranging from 22% [58] to 42% [59].

In 2013, excluding NMT, shares of private and public vehicles in total motorized passenger travel for KTM is about 63% and 37%, respectively (Fig. 3). In contrast, the corresponding figure for ROC is 20% (private vehicles) and the remaining 80% (public vehicles). The difference in KTM and ROC is mainly due to high share of motor cycles in total motor vehicle mix in KTM. In recent years, mobility by motor cycles has grown faster than any other transport modes and this is likely to continue in the future mainly due to its affordability and efficient short-distance traveling mode. Within total motorized public pkm traveled by vehicle types, motor cycle dominates in KTM and bus dominates in ROC.

While passenger travel patterns are more closely related to personal wealth and lifestyle changes, freight transport activities are closely connected to overall economic activity. In the case of road freight transport, 11 billion tkm of total freight demand is estimated for Nepal in 2013. Similar to road passenger transport, 0.3 tkm per US\$ of GDP (in PPP, constant 2005 prices), a measure of freight transport performance, is one of the lowest, compared to Australia (12.1 in 2008), Japan (4.2 in 2009), South Korea (0.5 in 2009), Pakistan (17.9 in 2005), United Kingdom (3.4 in 2009) and the USA (6.9 in 2006) [40]. It is interesting to note that almost one-fourth of total freight movement is concentrated in KTM indicating its importance in country's economic activity. Within vehicle categories, trucks (50%) dominate the total freight demand, followed by tractors (33%), pick-ups (10%), mini trucks (4%) and others (3%) in the country (Table 5). Besides trucks, pick-ups (22%) and mini trucks (12%) dominate the total freight demand in KTM, while it is tractors (42%) in ROC.

In recent years, air passenger volumes, measured in number of passengers, grew rapidly in Nepal. During the post-conflict era (2006–2012), the number of international and domestic

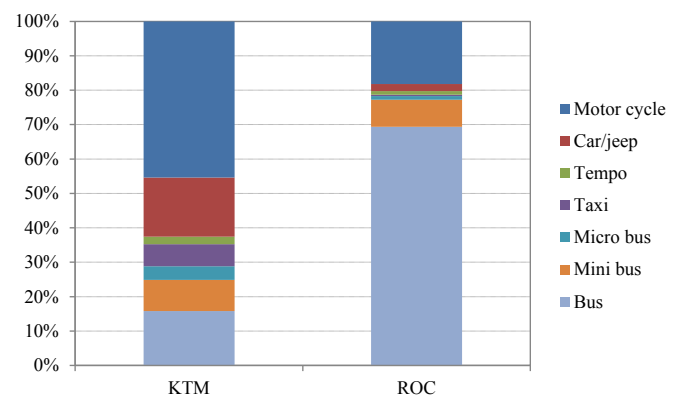


Fig. 3. Estimated shares of motorized road passenger travel demand by vehicle mode, 2013.

passengers grew strong at 13.3% and 10.1% per year, respectively (Fig. 4). This is evident from an increase in aircraft movements. For example, international aircraft movements more than doubled from 11057 in 2006 to 23320 in 2012, while domestic aircraft movement increased from 61291 in 2006 to 70877 in 2012, an increase of 16%. Increase in number of tourists visiting Nepal and the labor market traffic for the middle-east and Asia are most likely to be major factors responsible for rapid growth in air passenger volume in the country. Unlike air passengers, air freight movement has weak growth in recent years. During 2006–2012, domestic air freight grew by 0.5% per year, while international air freight grew only by 1.2% per year.

4.2. Energy analysis

Energy use in Nepal has grown rapidly over the past few years and transport is the major commercial energy-consuming sector. Energy used in transport remains heavily dominated by petroleum fuels. Between 2006 and 2013, gasoline sales increased by 2.7 times and diesel sales increased by 2.4 times, while ATF sales increased by 80% (Fig. 5). Prices of transport fuels also increased steadily over the last seven years. During the same period, price of gasoline, diesel and ATF doubled. Nepal has no significant reserves of fossil-fuel resources and all petroleum products are imported from India. Relatively higher price of petroleum fuels in 2008 is mainly due to price hike in India.

Transport sector used 816 ktoe of energy in 2013, more than two-thirds of country's total petroleum products use, including LPG which is mainly used for cooking. Gasoline, diesel and ATF combined contributed about 98% of total transport energy use. ATF, 99% of gasoline and 87% of diesel are used for transport sector in the

Table 5

Estimated motorized and non-motorized road travel demand by region and road classification, 2013.

Road transport classification			KTM	ROC		
Passenger (million pkm)						
Motorized	Private	Motor cycle	3565	5852		
		Car/jeep	1350	649		
		Tempo (3 wheeler)	171	345		
	Public	Taxi	501	125		
		Micro bus	314	346		
		Mini bus	707	2521		
		Bus	1242	22296		
		Non-motorized	Private	Walk	1656	9208
				Bicycle	135	3663
Freight (million tkm)						
Motorized	Public	Truck	1512	4012		
		Mini truck	289	207		
		Pick up	556	531		
		Tractor	26	3574		
		Others	122	193		

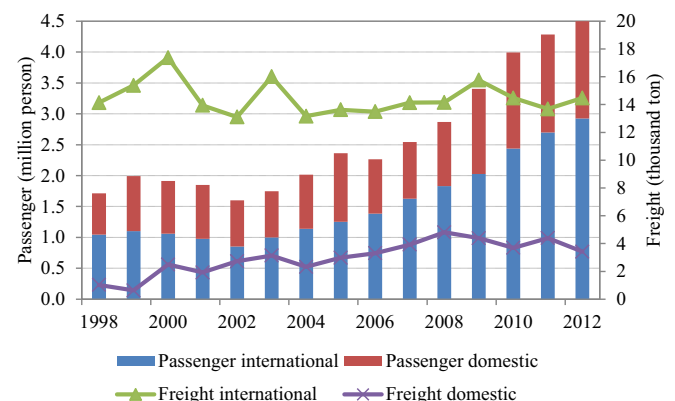


Fig. 4. Air transport activity at Tribhuvan international airport, 2007–2012 [9].

country. Electricity, mainly used for charging batteries for tempos, and LPG play a very small role in transport sector energy-mix. Region-wise, KTM accounts for substantial amount transport energy use. In 2013, e.g., about one-third of country's total energy use in road transport is consumed in KTM. Likewise, roughly half of country's total gasoline and one-fifth of total diesel are consumed in KTM in 2013 (Table 6).

In terms of energy use by vehicle ownership, private passenger vehicles consume 59% of KTM's total road passenger transport energy use. This is mainly due to operation of large number of cars/jeep and motor cycles (Fig. 6a). In contrast, 55% of total passenger energy use in ROC is consumed by public passenger vehicles dominated by mini buses and buses. Gasoline is the major fuel in road passenger-private transport while diesel is the major fuel in road passenger-public and freight transport. In the road freight transport, roughly one-fourth of total energy use, all diesel fuel, is consumed in KTM. Within freight road transport, trucks and pickups combined consumed 77% of total energy use in KTM, while trucks and tractors combined consumed 85% of total energy use in ROC (Fig. 6b). Small number of electric- and LPG- tempos, and LPG micro buses are operated in the country but the amount of energy use from these vehicles is insignificant. Fuel use in air transport, ATF, accounted for 14% of total transport energy use in 2013.

4.3. Emission analysis

Transport sector, mainly road, is responsible for deteriorating air quality in country's many urban areas, including KTM. In this section, transport sector's contribution to local and global air pollution in Nepal is presented. The estimated local air pollutants for road transport include CO, HC, NO_x, SO₂, PM₁₀ and PAH. In addition, a major greenhouse gas (CO₂) from road- and air-transport is estimated.

In 2013, road transport emitted 67 kt of CO, 19 kt of NO_x, 11 kt of HC, 11 kt of PAH, 4 kt of SO₂ and 3 kt PM₁₀ in the country (Table 7). In absolute values, clearly, KTM dominates in country's total emissions. This is mainly due to most of Nepal's economic activities, excluding agriculture, and rapid increase in urban population occurring in the capital. For example, KTM accounted for one-third of country's total CO, HC and PAH and more than one-fourth of total NO_x, PM₁₀ and SO₂ emissions in 2013 (Fig. 7). In KTM, private passenger transport is responsible for most of CO and HC emissions, while freight transport is responsible for most of NO_x, SO₂, PM₁₀ and PAH emissions. With the exception of SO₂ and PAH emissions, the share of public passenger transport in KTM's total emissions is quite low. In contrast, freight transport is responsible for most of NO_x, SO₂, PM₁₀ and PAH emissions, while private passenger transport is responsible for most of CO and HC emissions in ROC.

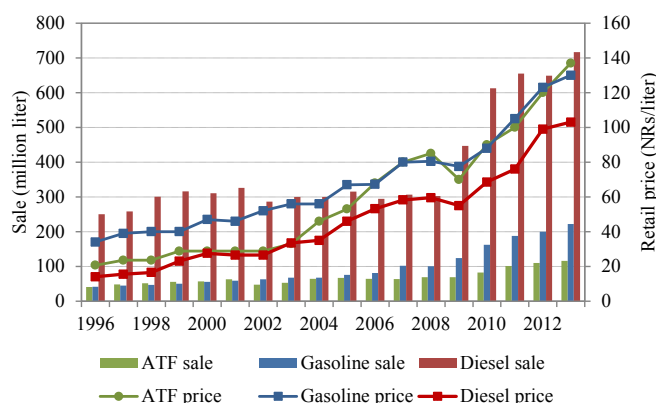


Fig. 5. Sale and retail price of selected petroleum products, 1996–2013 [7].

Most of these estimated local pollutants in ROC are likely to be concentrated along the east–west highway in tarai region and in urban cities in the country, including Pokhara, Biratnagar and Birgunj. Most of the rural hill- and mountain-regions in the country are likely to have less problem associated with transport related local air pollutants mainly because these regions lack access to motorized transport.

Most existing studies, including this paper, use bottom-up approach in estimating local air pollutants on an annual basis using activity level data and emission factors for road transport. Due to different dataset and sources on emission factors and energy use in transport, emission inventory of local air pollutants in KTM often vary widely (Table 7). For example, although estimated year is different, emissions of CO [14], HC [25] and PM₁₀ [17] are most likely overestimated, while CO emissions [25] are most likely underestimated. In fact, these studies use conventional tier-1 emission factors, with fuel as the activity indicator, in estimating these local air pollutants. This study uses tier-2 emissions factors reflecting more recent (Euro I) emission standards in the country.

In the case of road CO₂ emission, freight transport is responsible for 55%, followed by private passenger transport (24%) and public passenger transport (21%) in the country (Table 8). Although air transport is responsible for emissions of 292 kt of CO₂ in 2013, only about 10% of emissions are produced during airport ground level operations and during the landing/take-off cycle [57]. Region-wise, KTM is responsible for 31% and ROC is responsible for remaining 69% of total CO₂ emissions from road transport. In KTM, freight transport accounted for 38% of total CO₂ emissions, followed by private (37%) and public (24%) passenger transport, while in ROC, it is freight transport (62%), followed by public (20%) and private (18%) passenger transport (Fig. 8).

4.4. Mobility, energy use and air pollution: how Nepal compares to other countries?

High levels of motorization have close correlation with high levels of income. Both per capita GDP and number of cars per 1000 people in the developing countries are much lower compared to industrialized nations. For example, in 2010, the number of passenger cars per 1000 people is 627 in USA, 517 in Germany, 457 in UK, 453 in Japan, 276 in South Korea, 44 in China, 20 in Sri Lanka, 13 in Pakistan, 12 in India, 4 in Nepal and 2 in Bangladesh [40]. Within the region (i.e., South Asia and China), Nepal has the second lowest level of motorization after Bangladesh, while Bhutan and China have the highest level of motorization (Fig. 9). Many factors influence low level of private motorization in Nepal, notably, low level of income and real wage, fuel price and lack of road networks due to country's rugged topography. For example, the ratio of fuel price to per capita income in Nepal is among the highest in the world. In 2010, gasoline pump price per capita of GDP in Nepal is 60 times as high as in the USA, 21 times as high as in Japan, 7 times as high as in China and 3 times as high as in India. Furthermore, high level of crowding in urban centers, including in KTM, undermined the practicality of private motorization use mainly due to increasing costs of vehicles movement and storage, congestion, increasing feasibility of NMT modes and viability of public transport. For example, it takes only about 13 min and walk 1.1 km for people living in urban KTM to reach nearest bus stop [5]. However, as developing countries become more prosperous, motorization use is also expected to increase significantly. The trend line shown in Fig. 9 illustrates roughly the average number of passenger vehicles per 1000 people relative to GDP per capita. Reasons for high level of passenger cars per 1000 people relative to GDP per capita over the past decade in Afghanistan and Bhutan include low population density, and lack of railways and air transport system.

Table 6
Estimated transport energy use by region and fuel type, 2013.

Region	Road transport type			Fuel type				
	Passenger public	Passenger private	Freight	Gasoline	Diesel	LPG	Electricity	ATF
	(ktoe)	(ktoe)	(ktoe)	(ml)	(ml)	(ton)	(MWh)	(ml)
Kathmandu valley	58	83	84	109	159	303	7030	116 ^a
Rest of the country	104	87	301	110	462	1014	3915	

^a National total.

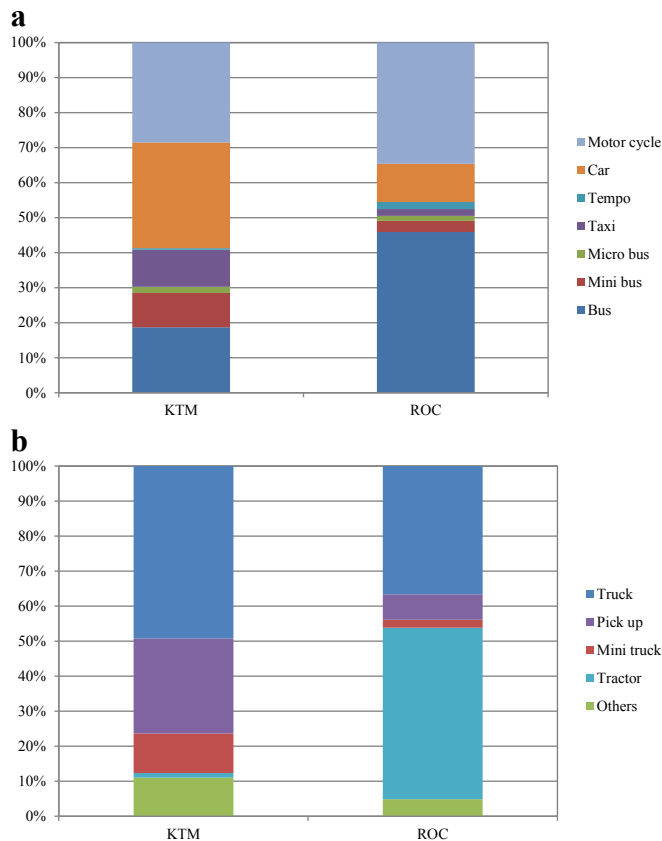


Fig. 6. a. Share of motorized road passenger transport energy use by vehicle mode, 2013. b. Share of motorized road freight transport diesel use by vehicle mode, 2013.

Similarly, in per capita terms, there is a positive correlation between income and energy use by road transport in many developing countries in the region. The trend line shown in Fig. 10 illustrates roughly the average amount of energy use by road transport per capita relative to GDP per capita. On average, countries above the trend line are more energy intensive in road transport (e.g., Pakistan and Sri Lanka) compared to those below the line. However, two countries (Nepal and China) are situated slightly above the trend line particularly in recent years, indicating moving toward more energy intensive road transport. In contrast, India is moving toward less energy intensive road transport in

recent years. Several factors can influence the position of a country or region relative to the trend line including socio-economic development, transport mode and intensity, and fuel mix. In the case of Nepal, road dominates all transport, and between 2003 and 2010, in per capita terms, road sector energy use increased by more than 111% while GDP increased by only 18%.

Despite this quite high growth in per capita road sector energy use over the past years, Nepal's level of energy use for road remains one of the lowest in the world. For comparison, Nepal's road energy use index of 0.011 is at the bottom of rankings compared to other developing and developed countries, and selected regions of the world, reflecting country's low consumption of energy use per capita for road (Fig. 11). For example, country's energy use per capita for road transport in 2010 is where S. Korea was in 1977, Pakistan was in 1978, India was in 1987 and China was in 1991.

5. Conclusions

The paper has estimated current level of travel demand and associated energy consumption and emissions for KTM and Nepal using ASIF framework. The analysis has provided general understanding of these issues in road transport by its different travel and vehicle modes and in air transport by passenger types.

Roads and air are the main transport modes in Nepal. However, their infrastructure and services are less developed and poor. Nepal's road density measured in both km per people and km per land area is one of the lowest in the South Asian region. Besides, only 48% of strategic road network (5585 km) and 3% of local road network (1528 km) in the country are paved roads and 91% of them are concentrated in the tarai and hill areas. Air services are limited to selected cities and few remote areas which are inaccessible by roads.

Within roads, both motorized and non-motorized travel is significant in Nepal. Their estimated activity levels and corresponding energy use and emission of air pollutants vary widely across different transport modes. First time these estimates are made using number of different transport modes and fuel types both for KTM and for the country as a whole. In 2013, Nepal's road passenger travel is estimated at 54.7 billion pkm, or about 1996 km per person. KTM dominates this travel demand, accounting for 18% of total. About three-fourth of country's total passenger travel is met by motorized vehicles and the remaining one-third is met by non-motorized transport. In the same year, shares of private and public vehicles in KTM's total motorized passenger travel are about 70% and 30%, respectively. In contrast, it is only 25% for private vehicles

Table 7
Estimated emissions of local air pollutants from energy use in road transport, 2013 (kt).

	KTM ^a						ROC					
	CO	HC	NO _x	SO ₂	PM ₁₀	PAH ^b	CO	HC	NO _x	SO ₂	PM ₁₀	PAH ^b
Passenger public	1.4	0.2	1.1	0.3	0.2	0.7	1.6	0.3	2.9	0.6	0.4	0.5
Passenger private	21.4	3.2	1.7	0.1	0.2	0.4	33.1	5.0	2.6	0.1	0.2	0.2
Freight	2.1	0.3	2.5	0.6	0.4	2.5	7.1	1.6	7.8	2.3	1.4	6.9

^a For comparison, in Kathmandu valley [14], estimated 37 kt of CO, 11 kt of HC, 2.1 kt of NO_x, 0.4 kt of SO₂ and 0.6 kt of PM₁₀ emissions in 2004 [25]; estimated 14.3 kt of CO, 9.2 kt of HC, 2.1 kt of NO_x, 0.2 kt of SO₂ and 0.5 kt of PM₁₀ emissions, while [17] estimated 20.1 kt of CO, 5.1 kt of HC, 11.4 kt of NO_x, 0.4 kt of SO₂ and 85.2 kt of PM₁₀ emissions in 2005.

^b Values reported in ton.

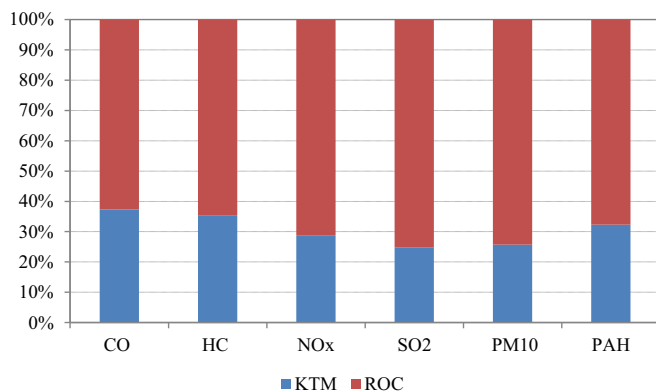


Fig. 7. Share of local air pollutants by region, 2013.

Table 8

CO₂ emissions from energy use in transport sector, 2013 (kt).

	KTM	ROC	Total
Road			
Passenger public	165	299	464
Passenger private	255	276	531
Freight	262	933	1195
Air			292

and 75% for public vehicles in ROC. In the case of freight transport, country's total demand is estimated at 11 billion tkm, or 0.3 tkm per GDP. Almost 23% of this freight movement is concentrated in the valley. Although number of passengers traveling by air has increased in recent years, their share in total travel demand remains minimal.

In 2013, two-thirds of country's total petroleum products, including LPG, are consumed by transport sector, mainly road. Nepal's road energy use per capita, about 26 kgoe, is one of the lowest in the world. Diesel and gasoline, all imported from India, are the main road transport fuels. Almost half of gasoline and one-fifth of diesel are consumed in KTM. Large number of private passenger vehicles, mainly motor cycles and cars, and taxis are responsible for all gasoline consumption, while public passenger and freight transport is responsible for most of diesel consumption. Small number of battery-operated tempos, and LPG-run tempos and micro buses are operated in the country. However, energy use by these vehicles is insignificant. ATF, fuel used in air transport, is accounted for 14% of total transport energy use in 2013.

Increasing petroleum based travel activities is mainly responsible for deteriorating air quality, traffic congestion and road accidents in Nepal, especially in KTM. By road transport modes, private passenger vehicles are responsible for most of CO and HC emissions,

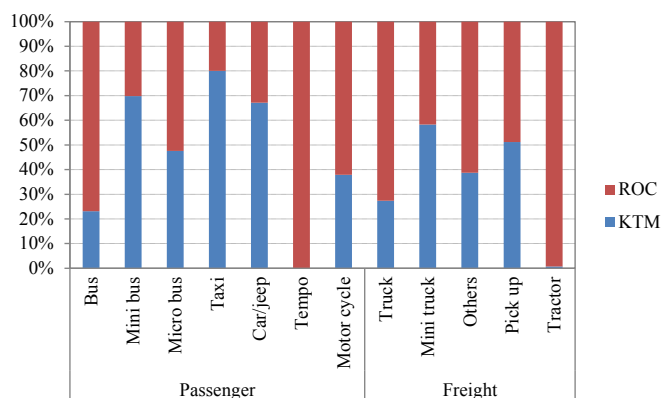
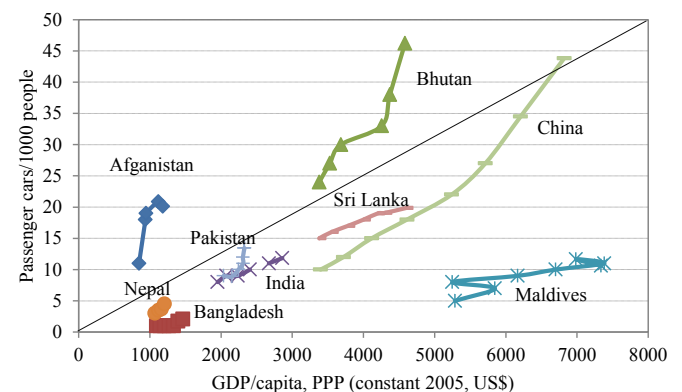
Fig. 8. CO₂ emissions from road transport by vehicle type, 2013.

Fig. 9. Relation between income and road motorization in South Asia and China, 2003–2010 [40].

while freight vehicles are responsible for most of NO_x, SO₂, PM₁₀ and PAH emissions in KTM. Most of these local air pollutants in ROC are most likely concentrated along the east–west highway in tarai region and in urban cities. Although transport sector (road and air) is responsible for half of country's total energy-related CO₂ emissions, its contribution to global transport-related CO₂ emissions, about 0.03% in 2012, is insignificant. Moreover, in per capita terms, Nepal's 0.12 tCO₂ is one of the lowest in the world. It is likely that Nepal's share in global energy-related CO₂ emissions will remain infinitely small in foreseeable future. This suggests that research studies and government actions on transport related CO₂ emissions in the country should focus more on adaption and less on mitigation.

Furthermore, without any sound transport plans and policies, increasing rate of motorization and urbanization combined with limited transport infrastructure development will likely to further deteriorate air quality and increase traffic congestion and road accidents in the country. It will also put pressure on government's foreign currency reserves from importing ever increasing petroleum products in the country. These issues will have negative impacts on economic growth and sustainable transport development in the country. Further research works that explores these issues at the regional level for future years would be worthwhile.

There are some limitations of our research. First, we have used values for annual VKT from the past literature in estimating travel demand. This may lead to over-estimates of travel demand because of increasing road congestion that may have lowered the VKT in recent years. Second, we assumed all vehicles registered in KTM operate in KTM and all vehicles registered in ROC operate in ROC. However, in reality, some of these vehicles, mainly buses, mini buses and trucks, operate between KTM and ROC. This may lead to

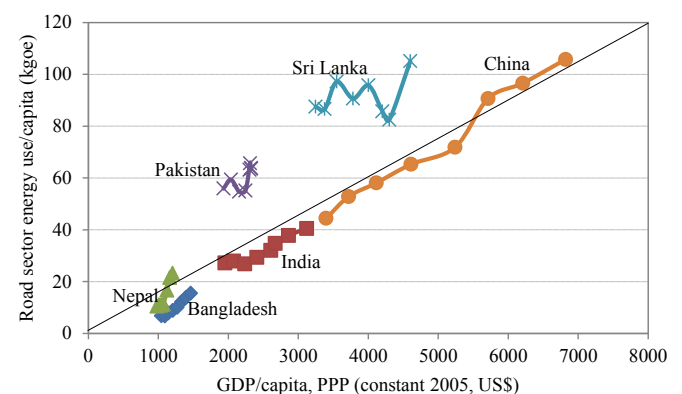


Fig. 10. Relation between income and energy use by road transport in South Asia and China, 2003–2010 [40].

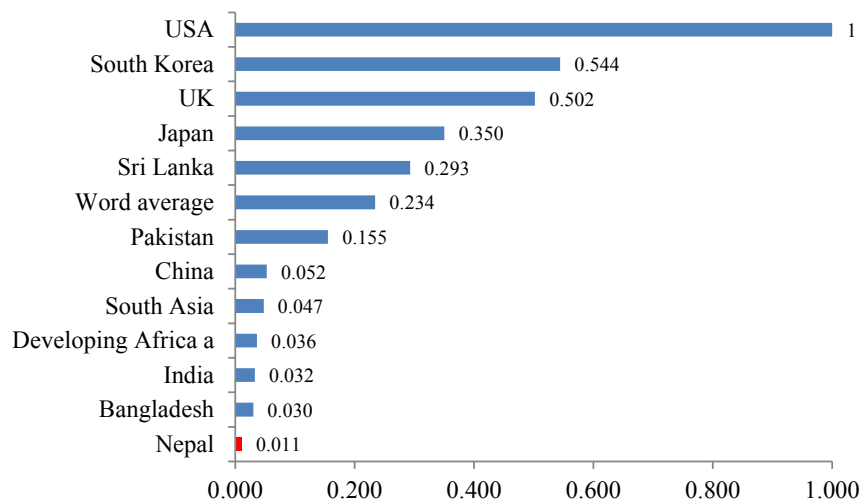


Fig. 11. Road sector energy use index of selected countries and regions of the world in 2010. ^a Developing Africa includes sub-Saharan developing Africa. Road sector energy use index is calculated as the arithmetic mean of each index created from four indicators, namely, per capita gasoline consumption, per capita diesel consumption, per capita other energy consumption and share of road sector energy consumption as % total energy consumption. Each indicator is calculated using the formula: Indicator = (actual value – minimum)/(maximum value – minimum value).

under- or over-estimates of fuel consumption and emissions in KTM and ROC. However, in spite of these limitations, we believe that results from our study can be used as the base year dataset in forecasting travel demand and associated energy consumption and emissions in the future.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.energy.2014.03.041>.

Table A.1

Road transport statistics and parameters used in the study [5,6,14,18,31,37,60–62].

Energy source		No. of registered vehicles ^a		OF ^b	VKT/yr (km)	LF ^c	EI (Km/l)
		KTM	ROC				
Motorized passenger							
Motor cycle	Gasoline	548430	658831	0.5 (0.6)	10000	1.3	53.8
Car/jeep	Multiple	82650	44085	0.4	16200	2.5	13.3
Tempo	Multiple	2494	5016	0.29 (0.58)	32340	7.3	12.8 ^e
Taxi	Gasoline	7000	5000	0.8 (0.5)	33413	2.5	11.7
Micro bus	Multiple	1298	1338	0.55	37125	12.3	8.8
Mini bus	Diesel	6705	1279	0.39 (0.45)	37125 (46406) ^f	26.4	5.4
Bus	Diesel	7532	22606	0.45 (0.42)	39600 (49500) ^f	33(50)	3.6
Motorized freight							
Truck	Diesel	12070	25574	0.39	39160 (48950)	8.2	3.9
Mini truck	Diesel	4023	1300	0.37 (0.45)	39160 (48950)	4.9	5.4
Pick up	Diesel	9525	8646	0.68 (0.7)	21054	4.1	5.2
Tractor	Diesel	2892	80209	0.11 (0.5)	21054	4.1	5.2
Others	Diesel	3409	9139	0.56 (0.39)	15664	4.1	2.8
Non-motorized passenger ^d							
Bicycle		64317	1744301	0.7	3000	1	
Walking		2693173	24553744	0.41 (0.25)	1500	1	

Figures in the parenthesis are for ROC. The following conversion factors are used: gasoline (0.792 kgoe/l), diesel (0.874 kgoe/l), ATF (0.842 kgoe/l), LPG (1.102 kgoe/kg) and electricity (85.985 kgoe/MWh).

^a Available official classification of motor vehicles is very broad in Nepal. We disaggregated motor vehicles by vehicle type, ownership and fuel type using various assumptions, past studies and available statistics as follows: 80% of bus and mini bus are public in KTM and ROC; for micro bus, we assume 3% diesel, 82% gasoline and 15% LPG in KTM and ROC; in KTM, 8% of total car/jeep is taxi, 48% is car and 44% jeep; in ROC it is 15% taxi, 41% car and 44% jeep; 50% of jeep is gasoline and 50% is diesel both in KTM and ROC; 40% of country's mini bus/mini truck is mini truck; we assume no gasoline and diesel tempo operate in KTM; 78% of heavy freight vehicles (crane, dozer, excavator, truck) is truck in KTM and it is 74% in ROC.

^b OF (ratio of operational to registered vehicle) are relatively high in some of the vehicle categories due to large number of new vehicles.

^c LF (occupancy) is the average number of passengers for passenger vehicles and average weight (in ton) for freight vehicles. Figure reported under tempo is for electric, diesel and LPG and it is 2.5 for gasoline LPG.

^d Data for non-motorized vehicles reported under the number of registered vehicles are in number for bicycles and population for walking. Bicycles do not require registration in Nepal.

^e Figure reported is for gasoline tempo. For diesel tempo it is 13.8 km/l, for LPG tempo it is in km/cylinder and each cylinder weighs 14.2 kg, and for electric tempo it is in MWh/vehicle per year based on 27 days per month of operation.

^f Figures reported are for public bus and mini bus. For private bus and mini bus, they are 15840 km and 14850 km respectively in KTM and ROC.

Table A.2Emission factor of local air pollutants^a for motorized road vehicles (g/km) [63–65].

		CO	HC	NO _x	SO ₂	PM ₁₀	PAH ^b
Motorized passenger							
Motor cycle	Gasoline	7.17	1.08	0.49	0.01	0.04	<0.00
Car	Gasoline	3.93	0.65	0.17	0.05	0.01	0.30
Jeep/van	Gasoline	3.93	0.65	0.21	0.05	0.01	0.46
	Diesel	0.45	0.05	1.48	0.11	0.27	1.36
Tempo	Gasoline	4.47	1.57	0.61	0.03	0.01	0.27
	Diesel	2.09	0.16	0.69	0.06	0.35	0.80
	LPG	1.70	1.03	0.04	0.01	0.13	0.34
Taxi	Gasoline	4.23	0.56	0.75	0.05	0.01	0.14
Micro bus	Gasoline	8.82	0.61	0.21	0.71	0.01	0.46
	Diesel	0.58	0.14	2.12	1.42	0.48	8.27
	LPG	3.90	0.77	0.20	0.01	0.00	0.02
Mini bus	Diesel	0.66	0.19	2.12	1.42	0.48	8.27
Bus	Diesel	2.71	0.71	7.62	1.42	1.09	0.52
Motorized freight							
Truck	Diesel	7.33	0.60	9.75	1.42	1.31	4.03
Mini truck	Diesel	0.66	0.19	2.12	1.42	0.48	8.27
Pick up	Diesel	3.66	1.35	2.12	1.42	0.48	8.27
Tractor	Diesel	3.00	1.28	2.48	1.42	0.66	3.77
Others	Diesel	7.33	0.60	9.75	1.42	1.31	4.03

^a Emissions of CO₂ are estimated on the basis of the amount of fuel combusted and its carbon content. The following emission factors used: 69300 kg/TJ for gasoline, 74100 for diesel, 63100 for LPG and 71500 for ATF [57].

^b PAH is polycyclic aromatic hydrocarbons expressed in mg/km.

References

- [1] MOF. Estimates of expenditure for FY2013/14. Kathmandu: Ministry of Finance (MOF); 2014.
- [2] MOF. Estimates of expenditure for FY2010/11. MOF; 2010.
- [3] NRB. Quarterly economic bulletin, mid October 2013. Kathmandu: Nepal Rastra Bank (NRB); 2014.
- [4] MOF. Economic survey: FY2012/13. Kathmandu: MOF; 2013.
- [5] CBS. Nepal living standard survey: FY2010/11, statistical report, vol. 1. Kathmandu: Central Bureau of Statistics (CBS); 2011.
- [6] DOTM. Number of registered vehicles. Kathmandu: Department of Transportation and Management (DOTM); 2014.
- [7] NOC. Import and sales of petroleum products. Kathmandu: Nepal Oil Corporation (NOC); 2014. <http://www.nepalioil.com.np/Import-and-Sales/22/> [accessed February 2014].
- [8] MOCS. The report of high level committee on Nepal Oil Corporation reform under the chairmanship of honorary Bhim Acharya. Kathmandu: Ministry of Commerce and Supplies (MOCS); 2012 [in Nepali].
- [9] CAAN. Civil aviation report: 2013. Kathmandu: Civil Aviation Authority of Nepal (CAAN); 2014.
- [10] CEN ENPHO. Health impacts of Kathmandu's air pollution, report prepared for KEVA Secretariat under USAID/Nepal. Kathmandu: Clean Energy Nepal (CEN) and Environment and Public Health Organization (ENPHO); 2003. http://pdf.usaid.gov/pdf_docs/PNACW355.pdf [accessed August 2013].
- [11] Dhakal S. Implications of transportation policies on energy and environment in Kathmandu Valley, Nepal. Energy Policy 2003;31:1493–507.
- [12] Gautam C. Final report on action program on air quality management of Kathmandu Valley. Kathmandu: Ministry of Science, Technology and Environment (MOSTE); 2006.
- [13] ICIMOD. Kathmandu Valley environment outlook. Kathmandu: International Centre for Integrated Mountain Development (ICIMOD); 2007.
- [14] IGES. Urban transportation and the environment in Kathmandu Valley, Nepal: integrating global carbon concerns into local air pollution management. In: Dhakal S, editor. Kanagawa: Institute for Global Environmental Strategies (IGES); 2006.
- [15] MOSTE. Ambient air quality of Kathmandu Valley 2007. Kathmandu: MOSTE; 2007.
- [16] MOSTE. Environmental standards and collection of relevant policies. Kathmandu: MOSTE; 2010.
- [17] Pradhan BB, Dangol PM, Bhaunju RM, Pradhan S. Rapid urban assessment of air quality for Kathmandu, Nepal: summary. Kathmandu: ICIMOD; 2012.
- [18] Pradhan S, Ale BB, Amatya VB. Mitigation potential of greenhouse gas emission and implications on fuel consumption due to clean energy vehicles as public passenger transport in Kathmandu Valley of Nepal: a case study of trolley buses in Ring Road. Energy 2006;31:1748–60.
- [19] Shah J, Nagpal T. Urban air quality management strategies in Asia: Kathmandu Valley report. World Bank technical paper No. 378. Washington, DC: World Bank; 1997.
- [20] Shaky SR, Shrestha RM. Transport sector electrification in a hydropower resource rich developing country: energy security, environmental and climate change co-benefits. Energy Sustain Dev 2011;15:147–59.
- [21] Sharma CK. Overview of Nepal's energy sources and environment. Atmos Environ 1996;30:2717–20.
- [22] Sharma CK. Urban air quality of Kathmandu valley "Kingdom of Nepal". Atmos Environ 1997;31:2877–83.
- [23] Shrestha KL. Nepal's energy situation. Energy 1981;6:817–21.
- [24] Shrestha RM, Malla S. Air pollution from energy use in a developing country city: the case of Kathmandu Valley, Nepal. Energy 1996;21:785–94.
- [25] Shrestha RM, Rajbhandari S. Energy and environmental implications of carbon emission reduction targets: case of Kathmandu Valley, Nepal. Energy Policy 2010;38:4818–27.
- [26] Shrestha RM, Shaky SR. Benefits of low carbon development in a developing country: case of Nepal. Energy Econ 2012;34(Suppl. 3):S503–12.
- [27] Silveira S, Khatiwada D. Ethanol production and fuel substitution in Nepal: opportunity to promote sustainable development and climate change mitigation. Renew Sustain Energy Rev 2010;14:1644–52.
- [28] Shrestha SP, Shrestha K, Shrestha K. Carbon dioxide emissions by the transportation sector in Kathmandu Valley, Nepal. In: International conference on sustainable design, engineering, and construction (ICSDEC); 2012. pp. 90–7.
- [29] Ale BB, Shrestha SO. Introduction of hydrogen vehicles in Kathmandu Valley: a clean and sustainable way of transportation. Renew Energy 2009;34:1432–7.
- [30] Baral A, Parajuli R, Aryal B. Institutional responses to electric vehicle promotion in Nepal. Stud Nepali Hist Soc 2000;5:89–125.
- [31] Bhatta SD, Joshi DR. Are electric vehicles viable in Kathmandu? A cost-benefit perspective, prepared for Kathmandu electric vehicle alliance secretariat under USAID/Nepal. Cooperative agreement No: 367-A-00-02-00203-00 with PADCO, Inc. Washington, DC, http://pdf.usaid.gov/pdf_docs/PNACY587.pdf; 2004 [accessed August 2013].
- [32] USAID. The Kathmandu electric vehicle alliance (KEVA): final program performance report. Kathmandu: USAID; 2006.
- [33] Devkota B, Dudycha D, Andrey J. Planning for non-motorized travel in rural Nepal: a role for geographic information systems. J Transp Geogr 2012;24:282–91.
- [34] Dhungel DN, Pun SB. The Nepal-India water relationship: challenges. The Netherlands: Springer; 2009.
- [35] Practical Action. Ropes of hope. Kathmandu: Practical Action; 2012.
- [36] ADB. Proposed loan and Asian development fund grant Nepal: air transport capacity enhancement project, report and recommendation of the president to the board of directors, project no.: 34349. Manila: Asian Development Bank (ADB); 2009. <http://www.adb.org/sites/default/files/projdocs/2009/38349-NEP-RRP.pdf> [accessed August 2013].
- [37] ADB. Proposed loan, grant and administration of grant Nepal: Kathmandu sustainable urban transport project, report and recommendation of the president to the board of directors, project no.:44058. Manila: ADB; 2010. <http://www.adb.org/sites/default/files/projdocs/2010/44058-01-nep-rrp.pdf> [accessed August 2013].
- [38] World Bank. Nepal road sector assessment study. Washington, DC: World Bank; 2012. draft.
- [39] JICA. The study on Kathmandu Valley urban road development, final report, part A: master plan study. Kathmandu: Japan International Cooperation Agency (JICA); 1993.
- [40] World Bank. World development indicators. Washington, DC: World Bank; 2013.
- [41] CAAN. Civil aviation report: 2009–2010. Kathmandu: CAAN; 2010.
- [42] CBS. Statistical yearbook of Nepal: 2009. Kathmandu: CBS; 2010.
- [43] CBS. National population and housing census 2011: national report. Kathmandu: CBS; 2012.
- [44] DOLIDAR. Summary of rural road statistics: fiscal year 2011/12. Kathmandu: Department of Local Infrastructure Development and Agriculture Roads (DOLIDAR); 2012 [in Nepali]. <http://www.dolidar.gov.np/wp-content/uploads/2012/10/Rural-Road-Summary-2069.pdf> [accessed August 2013].
- [45] MOF. Economic survey: fiscal year 2011/12. Kathmandu: MOF; 2012.
- [46] WTTC. Travel and tourism: economic impact 2013 Nepal. London: World Travel and Tourism Council (WTTC); 2013.
- [47] NRB. The share of Kathmandu Valley in the national economy: survey report. Kathmandu: NRB; 2012.
- [48] DOR. Statistics of strategic road network (SSRN): 2011/12. Kathmandu: Department of Roads (DOR); 2012. http://www.dor.gov.np/documents/Road_Network_data.pdf [accessed August 2013].
- [49] Nepal Police. Annual accidental description. Kathmandu: Nepal Police; 2013. <http://traffic.nepalpolice.gov.np/traffic-updates/annually-accidental-descriptions.html> [accessed September 2013].
- [50] NPC. An approach paper to the thirteenth plan (FY2013/14–2015/16). Kathmandu: National Planning Commission (NPC); 2013.
- [51] TRB. Travel demand forecasting: parameters and techniques. Washington, D.C.: Transportation Research Board (TRB); 2012.
- [52] Ortuzar JD, Willumsen LG. Modeling transport. 4th ed. New York: Wiley and Sons; 2001.
- [53] Modelling transport (energy) demand and policies – an introduction. Energy Policy 2012;41:iii–xiv.
- [54] Schipper L, Marie-Lilliu C. Carbon-dioxide emissions from transport in IEA countries: recent lessons and long-term challenges. Paris: International Energy Agency (IEA); 1999.
- [55] Schipper L, Marie-Lilliu C, Gorham R. Flexing the link between transport and greenhouse gas emissions. Paris: IEA; 2000.

- [56] Chester VC, Horvath A. Environmental assessment of passenger transportation should include infrastructure and supply chains. *Environ Res Lett* 2009;4:1–8.
- [57] Intergovernmental Panel on Climate Change (IPCC). 2006 IPCC guidelines for national greenhouse gas inventories: volume 2 energy. In: Eggleston HS, Buendia L, Miwa K, Ngara T, Tanabe K, editors; 2006. Kanagawa: IGES.
- [58] Udas S. Public transport quality survey. Kathmandu: Clean Air Network Nepal (CANN) and Clean Energy Nepal (CAN); 2012.
- [59] CANN. Walking and cycling: a policy brief on non-motorized transport (NMT) system in Kathmandu valley. Kathmandu: CANN; 2012.
- [60] Arora S, Vyas A, Johnson LR. Projections of highway vehicle population, energy demand, and CO₂ emissions in India to 2040. *Nat Resour Forum* 2011;35:49–62.
- [61] NESS. Analysis of HMG policies and regulations affecting electrical vehicles: final report. Kathmandu: Nepal Environmental and Scientific Services (NESS); 2003. http://cleanairinitiative.org/portal/system/files/articles-59005_KEVA_Policy_Study.pdf [accessed September 2013].
- [62] WECS. Energy sector synopsis report 2010. Kathmandu: Water and Energy Commission Secretariat (WECS); 2010.
- [63] ARAI. Profiling for vehicular emissions: draft final report, report no.: ARAI/VSP-III/SP/RD/08-09/60. Pune: The Automotive Research Association of India (ARAI); 2009. http://www.cpcb.nic.in/Source_Profile_Vehicles.pdf [accessed August 2013].
- [64] EEA. EMEP/EEA emission inventory guidebook 2009, updated May 2012. Copenhagen: European Environment Agency (EEA); 2012. <http://www.eea.europa.eu/publications/emep-eea-emission-inventory-guidebook-2009> [accessed September 2013].
- [65] Ramachandra TV, Shwetmala. Emissions from India's transport sector: state-wise synthesis. *Atmos Environ* 2009;43:5510–7.